Copernicus Global Land operations
”Vegetation and Energy”
“CGLOPS-1”

Framework Service Contract N° 945120-IPR-2023

PRODUCT USER MANUAL

SOIL WATER INDEX (SWI) 0.1°
SOIL WATER INDEX 10 DAYS (SWI10) 0.1°
SWI TIME SERIES (SWI-TS) 0.1°

Issue I3.00

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LIST OF ACRONYMS

AD	Applicable Document
ASCAT	Advanced Scatterometer (Metop)
ATBD	Algorithm Theoretical Basis Document
CAL/VAL	Calibration and Validation Group of CEOS
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud-Optimized Geotiff
DGG	Discrete Global Grid
ECMWF	European Centre for Medium Range Weather Forecasting
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
GeoTIFF	Geographical Tagged Image File Format
GCOS	Global Climate Observing System
GIO	GMES Initial Operations
GLDAS	Global Land Data Assimilation System
GLWD	Global Lake and Wetlands Database
GMES	Global Monitoring for Environment and Security
HDF	Hierarchical Data Format
ISMN	International Soil Moisture Network
METOP	Meteorological Operational satellite
netCDF	Network Common Data Form
NRT	Near Real Time
PUM	Product User Manual
QAR	Quality Assessment Report
QFLAG	Quality Flag
RMSD	Root Mean Square Difference
SSF	Surface State Flag
SSM	Surface Soil Moisture
SWI	Soil Water Index
SWI10	Soil Water Index 10-day average
SWI-TS	Soil Water Index time series
TC	Triple Collocation
VR	Validation Report
WGS84	World Geodetic System 1984

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

The CLMS portfolio includes a global Soil Water Index (SWI) at 0.1° spatial resolution derived from the observations from the Advanced Scatterometer (ASCAT) on board the Meteorological Operational (Metop) satellite.

The Product User Manual (PUM) gives an overview of the characteristics of the global Soil Water Index (SWI) product at 0.1° resolution, the SWI calculated over a 10-days time period (SWI10) product, and the SWI time series (SWI-TS) product, in terms of algorithm, technical properties, and main validation results.

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1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The Product User Manual (PUM) is the primary document that users have to read before handling the products.

It gives an overview of the product characteristics, in terms of algorithm, technical characteristics (file format, product content, spatio-temporal characteristics, …), and a summary of the product validation results.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 gives an overview of the biophysical parameters, and presents the retrieval methodology with a discussion of error sources and products limitations.
- Chapter 3 provides an overall description of the products in term of file format, product content and other product characteristics.
- Chapter 4 summarizes the results of the product validation.
- Chapter 5 gives guidance on using the product, from T-value selection and masking to suitable applications.

The requirements and the expected performance are recalled in the Annex.

1.3 RELATED DOCUMENTS

1.3.1 Applicable document

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

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1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_SWI0.1	Algorithm Theoretical Basis Document presenting the detailed retrieval methodology of the SWI V4.0 product
GIOGL1_QAR_SWIV3-SWI10	Report describing the results of the scientific quality assessment of the SWI 0.1° V3.2 and SWI10 products
CGLOPS1_VR_SWI0.1-V4	Validation Report of Soil Water Index 0.1° Version 4.0

1.3.3 External documents

Document ID	Descriptor
ASCAT_SSM_NRT_12_5km _H29_PUM_v0_2	The Product User Manual of the H29 ASCAT SSM NRT 12.5km product.[1]
ASCAT_SSM_NRT_12_5km _H29_PVR_v0_1	The Product Validation report of the H29 ASCAT SSM NRT 12.5km product.[2]
GCOS-245	The 2022 GCOS ECVs Requirements

https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

[1] https://hsaf.meteoam.it/products/soil-moisture
[2] https://hsaf.meteoam.it/products/soil-moisture

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2 ALGORITHM

2.1 OVERVIEW

Soil moisture is a key parameter in numerous environmental studies including hydrology, meteorology, and agriculture. In addition to Surface Soil Moisture (SSM), which is the moisture content in the upper few centimetres of the soil, information on the moisture condition within the underlying soil profile is of interest for different applications. In a number of studies, the significance of root zone soil moisture has been addressed and different methods for estimating moisture content in the soil profile from surface measurements have been investigated. Soil moisture in plant root zone can be estimated by an infiltration model using information on surface soil moisture and soil characteristics.

The Soil Water index (SWI) product of the Copernicus Land Monitoring Service (CLMS) provides global daily information about moisture conditions in different soil depths. Near Real Time (NRT) Data prior to version 4.0 uses the EUMETSAT ASCAT-25km SSM product in orbit format (Bartalis et al. 2008) as input data. The SWI product version 4.0 uses the H29 Metop ASCAT Surface Soil Moisture Near Real Time 12.5 km product[3] as input data. The Archive of ASCAT SWI uses reprocessed SSM data from TU Wien. All SWI datasets include a quality flag indicating the availability of SSM measurements for the SWI calculations.

The SWI10 product aggregates the SWI data over 10-day-intervals to make combination with other Global Land products easier.

The SWI time series product (SWI-TS) contains the same data as the SWI product but is reformatted for easier time series access.

2.2 RETRIEVAL METHODOLOGY

2.2.1 Soil Water Index

The SWI algorithm, originally developed at Vienna University of Technology (TU Wien) and later improved by other research groups, uses an infiltration model describing the relation between surface soil moisture and profile soil moisture as a function of time. The algorithm is based on a two-layer water balance model proposed by (Wagner et al. 1999) to estimate profile soil moisture from SSM retrieved from scatterometer data. The following formula describes how the SSM moisture observations are transformed into the SWI:

[3] https://hsaf.meteoam.it/products/soil-moisture

$$SWI(t_n) = \frac{\sum_{i}^{n} SSM(t_i) e^{-\frac{t_n - t_i}{T}}}{\sum_{i}^{n} e^{-\frac{t_n - t_i}{T}}} \quad for\, t_i \leq t_n \qquad \text{(Eq.1)}$$

In which t_n is the observation time of the current measurement and t_i is the observation time of the previous measurements. These times are given in Julian days. In (Eq.1) all SSM observations made before t_n are summed up and exponentially weighted. The factor T determines how fast the weights become smaller and how strongly SSM observations taken in the past influence the current SWI. If e.g. the T-value is 5 and a SSM measurement was taken 10 days before t_n then this observation would receive the weight $e^{-\frac{10}{5}} = 0.135$. Using a T-value of 20 in the same case would increase the weight to $e^{-\frac{10}{20}} = 0.607$.

This formulation represents a simple two-layer water balance model, with the first layer as the soil surface layer accessible to C-band scatterometers and the second layer as the part of the profile which extends downwards from the bottom of the soil surface layer. The second layer is assumed to serve as a reservoir which is connected to the atmosphere via the first layer. The moisture conditions in the first layer, which is in direct contact with the atmosphere, are temporally highly dynamic. With increasing profile depth in the second layer, the temporal dynamic of the moisture conditions is decreasing, as the amount of water stored in the reservoir depends on the infiltration of water added to the first layer during precipitation events. Therefore, the water content of the reservoir layer is solely controlled by the past moisture conditions in the surface layer and thus the precipitation history. Here, more recent precipitation events have a stronger impact on the moisture conditions in the reservoir, which is accounted for by the exponential weighting function in Eq. 1. It is assumed that L is the depth of the reservoir layer (W_R) and C is an area-representative pseudo-diffusivity constant. The used T-value for the SWI calculation is T=L/C which means that a high T describes a deeper soil layer if the soil diffusivity is constant. It also means that the same T-value describes different depths for different soils. Figure 1 shows a simple sketch of the model. The link between the remotely sensed surface layer and the reservoir is described in more detail by (Ceballos et al. 2005).

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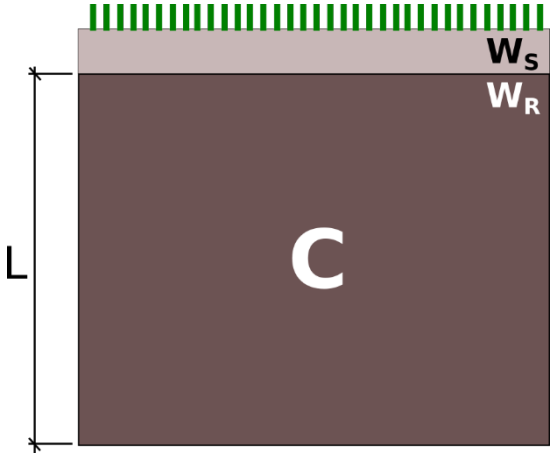


Figure 1: Sketch of the two layers model used in the SWI calculation. L is the reservoir depth, C the pseudo diffusivity constant and W_S and W_R denote the surface and reservoir layers.

For the SWI calculation provided in (Eq.1), time series of SSM measurements are required which may not be available at an operational level. A computational adaption of the SWI algorithm has been made based on a recursive formulation proposed by (Albergel et al. 2008). Using a recursive reformulation of the SWI, the calculation can be started at any time without the requisite for preceding SSM data. In this method, a gain factor is introduced that relates the past SWI measurements to the current measurements:

$$gain_T(t_n) = \frac{gain_T(t_{n-1})}{gain_T(t_{n-1}) + e^{-\frac{t_n - t_{n-1}}{T}}} \quad \text{(Eq. 2)}$$

$$SWI_T(t_n) = SWI_T(t_{n-1}) + gain_T(t_n)(SSM(t_n) - SWI_T(t_{n-1})) \quad \text{(Eq. 3)}$$

In (Eq.2) and (Eq.3), n refers to the latest, and $n-1$ to the previous *SSM* measurement. The SWI processing algorithm before product version 4.0 uses the near-real-time ASCAT-25km SSM product as input to generate daily global SWI images. In version 4.0, the input stream has been upgraded to use the H29 ASCAT SSM at 12.5km sampling.

SWI images are calculated for eight different T-values (1, 5, 10, 15, 20, 40, 60, 100) together with the respective quality flags. The SWI quality flag (QFLAG) is a parameter related to the number of available SSM measurements used for calculation of SWI_T at time *t*. The QFLAG describes the level of confidence that the SWI calculation has had sufficient data, of acceptable quality, within the

characteristic time length (T), to derive a meaningful SWI value. It is calculated by using a recursive formulation and expressed in percent:

$$qflag_T(t_i) = \begin{cases} 1 + qflag(t_{n-1}) \cdot e^{-\frac{t_n - t_{n-1}}{T}}, & if\ SSM\ at\ t_i\ is\ available\ (t_i = t_n) \\ qflag(t_{n-1}) \cdot e^{-\frac{t_i - t_n}{T}}, & if\ SSM\ at\ t_i\ is\ unavailable \end{cases} \quad \text{(Eq. 4)}$$

Figure 2 shows, for a point location, an annual evolution (2008) of the surface state flag (SSF) in comparison to surface soil moisture (SSM), soil water index (SWI) and the QFLAG for the characteristic time length T=20. The SSF is used to identify acceptable SSM values to be used in SWI calculations, i.e. SSM values obtained under non-frozen surface conditions. The SWI is initially masked, since the SSF shows frozen soil and the QFLAG is zero. With the onset of a continuous thaw during the beginning of May 2008, the SWI begins to change and the QFLAG increases. As soon as freeze-up occurs in October the SWI remains constant at the value of the last valid SSM measurement. After that the QFLAG drops at an exponential rate. As soon as the QFLAG is below 5%, the SWI is masked again.

In version 4.0, this SSF flag is not derived from ASCAT backscatter anymore, but is instead derived from meteorological data. Since this version also upgraded to use the H29 ASCAT SSM input-stream, the masking strategy was changed to use directly the advisory flag of the H29 product instead of the SSF to mask frozen soil conditions. The SSF flag is nonetheless kept in the product as auxiliary data for the users and for backwards compatibility.

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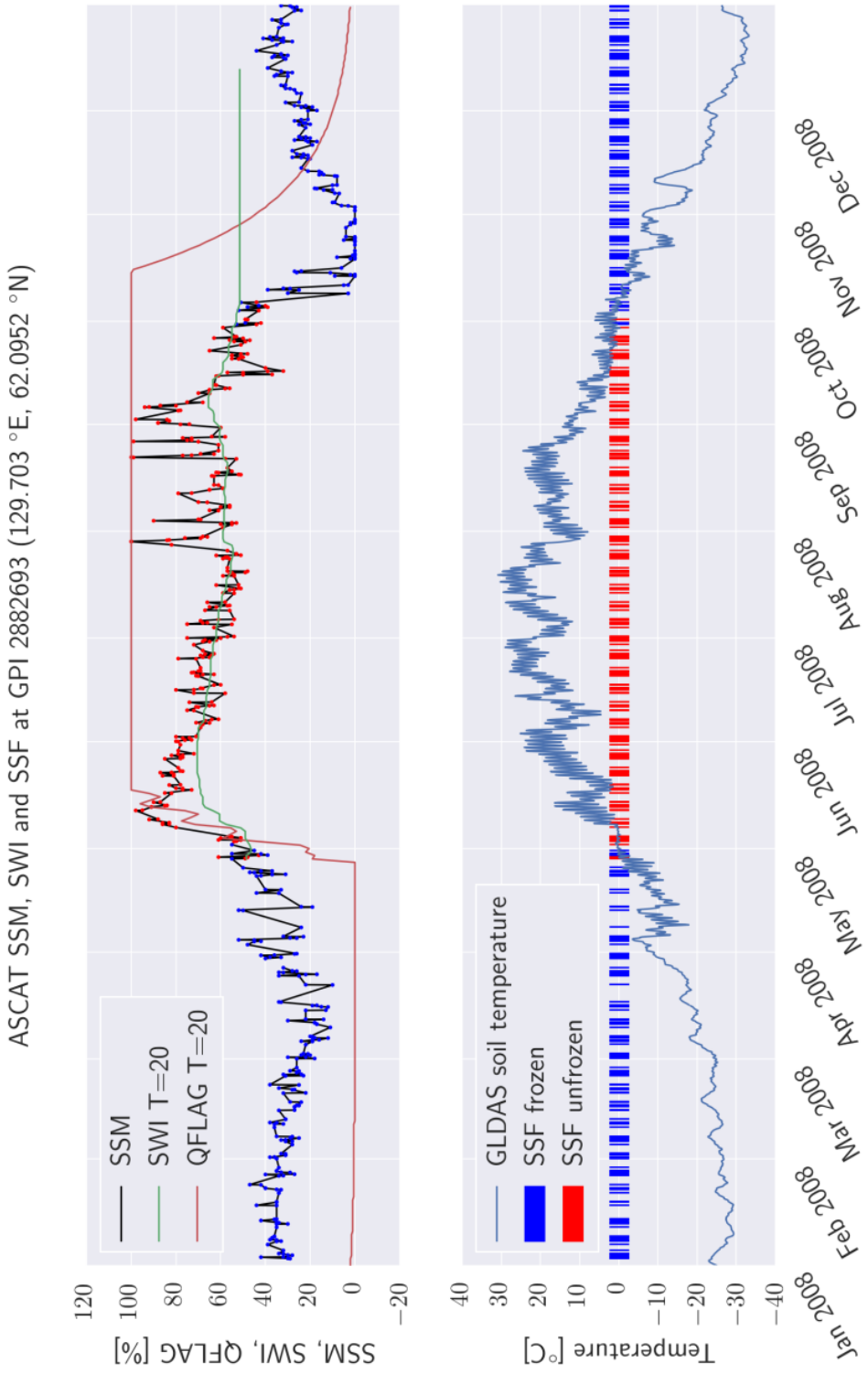


Figure 2: Example time series of SSF, SWI v3.2 and QFLAG

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A QFLAG of 100% indicates the ideal case that the SWI calculation has had sufficient data of acceptable quality and a QFLAG of 0% shows that, although a SWI calculation has been made, there were not enough adequate SSM measurements available to provide a confident SWI value.

2.2.2 Surface State Flag

Surface soil moisture (SSM) retrieval from scatterometer data has certain limitations: it cannot be retrieved when the surface is frozen or covered by snow, dense vegetation, or water. Whereas dense vegetation and water are almost static factors, the freeze/thaw cycle is dynamic.

The backscattered signal depends, among other factors, on the frozen state of the surface. When the soil surface freezes, dielectric properties of the soil change significantly which usually results in low backscatter values. As snow begins to fall and accumulates over the surface, backscatter signals may, depending on microwave frequency of the scatterometer, increase due to volume scattering. With increasing temperature in spring, snow begins to melt and water covers the surface of the snow pack which causes a sudden drop in backscatter. After the snow melting period, soil and vegetation begin to thaw and consequently backscatter rises again. Figure 3 shows this behaviour.

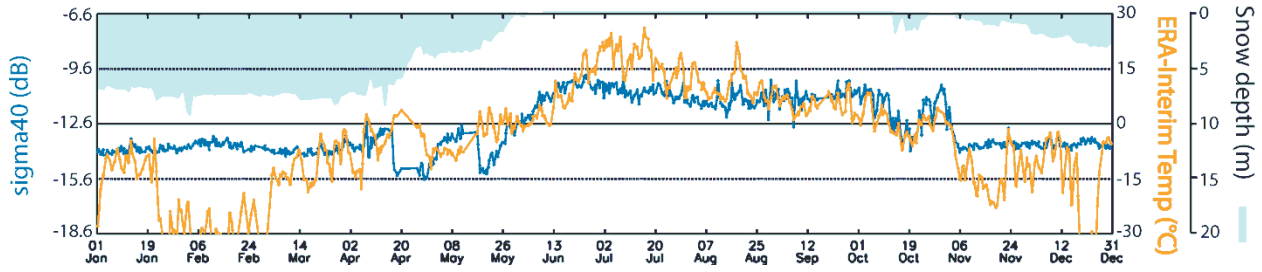


Figure 3: Comparison of ASCAT backscatter (sigma40) measurements, ERA Interim 2m Air Temperature and WMO Meteo snow depth data over a grid point at 66.5037°E, 66.6924°N in 2007

Table 1: Surface State Flag (SSF) values

SSF value	Detected Surface State
255	Could not be determined
0	Unknown
1	Unfrozen
2	Frozen
3	Temporary water on the surface

The frozen state of the surface is recorded within the surface state flag (SSF). Prior to version 4.0, the SSF is calculated in 2 steps from backscatter observations. Initially, the algorithm uses the behaviour

of backscatter measurements in response to temperature changes to find freeze/thaw parameters that define the freezing characteristics of each location. In a second step, two decision trees use these parameters to find the freeze/thaw state of every soil moisture observation. Table 1 shows the possible values of the surface state flag. If the SSF has the values 2 or 3, the soil moisture measurements are not usable for SWI calculation, and the SWI is kept constant at the level it had before freezing.

In product version 4.0, the advisory flag of the H29 ASCAT SSM product is used directly to mask SSM for frozen soil conditions. The SSF flag is only kept as auxiliary data for the user and for backwards compatibility. In this version the SSF flag is calculated from meteorological forecast data, similar to the advisory flag of the H29 product. ECMWF provides forecasts for skin temperature and snow depth every 2 hours. For each forecast within the 24-hour time-frame of the SWI product, we create 12 binary masks where skin temperature is below 0° C or snow depth is above 0 m. Then the SSF flag is set to "Frozen" for a certain point if the majority of these 12 masks indicate frozen surface conditions at that point. If no meteorological forecast is available for that day the flag is set to "Unknown". Over areas where no meteorological forecasts for land surface variables are available, such as the ocean, the flag is set to "Could not be determined". The remaining flag "Temporary water on the surface" is deprecated, but is still kept in the data format to allow for backwards compatibility.

2.2.3 10-daily Soil Water Index (SWI10)

The SWI10 product takes the SWI data as input and calculates the mean of all valid observations for each pixel. The periods over which the averaging is done are defined as follows: days 1 to 10, days 11 to 20, days 21 to end of month for each month of the year. The mean calculation is done for all eight T-values of SWI and QFLAG. Only values where the SSF shows the value "unfrozen" are considered for the calculation of the mean. For each pixel and T-value, a variable called VOBS is introduced that contains the percentage of valid observations in the averaging period.

2.2.4 Soil Water Index Time Series (SWI-TS)

The SWI-TS product is the SWI daily product in a time series format which makes it easier to perform local, time series-based analysis without having to download and convert data for the whole globe. The time series data is provided on a Discrete Global Grid (DGG) with a grid point spacing of 12.5 km. This grid contains a total of 3,264,391 points of which 839,826 are over land areas. This grid has the advantage of reducing the data volume of the product while not losing any information. This is the case because the SWI image product is oversampled to a 0.1° grid from this DGG and would contain 1,814,400 points over land.

This product is provided every six months and is updated in July and January.

2.3 LIMITATIONS OF THE PRODUCT

The SWI retrieval method does not account for soil texture, which determines the relation between the T-value and soil depth (de Lange et al. 2008). Therefore, eight T-values are provided to give more possibility to the users for selecting the best matching data.

Evapotranspiration is not considered when calculating the SWI which can lead to artificially high root zone soil moisture estimates when precipitation time and satellite observation time coincide and the rainfall evaporates instead of filtrating into deeper soil layers. In areas with a high ground water table, the SWI method does not work particularly for the deeper layers affected by the ground water.

In addition, the SWI algorithm cannot be applied in regions for which a surface soil moisture extraction from ASCAT C-band scatterometer data is not meaningful. This especially applies to high altitude permafrost regions, glaciers, ice caps, dense forests (e.g. tropical rainforest) and deserts.

The SWI product is based on a SSM product, which uses a daily climatology per grid point for accounting for the influence of vegetation on the backscattered signal. This means that the SSM product assumes the same vegetation development each year. If the actual vegetation state in an area is significantly different from the average, this can lead to biases in the retrieved soil moisture values.

See Section 5.1 for more details on how to choose T-values and how to use auxiliary data to identify regions where SWI retrievals can be unreliable.

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3 PRODUCT DESCRIPTION

3.1 IMAGE PRODUCTS

3.1.1 SWI

3.1.1.1 Overview

The Near-Real Time SWI product contains daily synthesis of Soil Water Index derived from H29 Metop ASCAT Surface Soil Moisture Near Real Time 12.5 km products. SWI products prior to version 4.0 used ASCAT SSM data at 25 km resolution. The values are given on a global 0.1° regular grid of 1800 lines and 3600 columns. For each node (pixel), 8 SWI values and their corresponding 8 quality flags, related to different T-values of characteristic time length (T=1, T=5, T=10, T=15, T=20, T=40, T=60 and T=100) as well as the Surface State Flag are available.

The products are saved in netCDF format according to the NetCDF Climate and Forecast (CF) Metadata Convention (CF conventions) (Eaton et al. 2011). The CF conventions define multiple possible representations of geographic and time series data.

3.1.1.2 File format

The SWI products are provided in netCDF4 format. The filename respects the following convention:

c_gls_SWI_YYYYMMDDHHMM_GLOBE_ASCAT_VX.Y.Z.nc

In all filenames, the following placeholders will be replaced by actual values:

- YYYYMMDD is the date of the product
- HHMM is set to 1200 to indicate that the product is representative of 12:00 UTC
- VX.Y.Z is the version in which X is the major version, Y the minor version and Z the run number. Table 2 below explains the recommended approach for handling updates to the versioning.

Table 2: Explanation in version numbering and recommendations for using efficiently the products

Versions	Differences	Recommendations
Major	Significant change to the algorithm.	Do not mix various major versions in the same applications, unless it is otherwise stated.
Minor	Minor changes in the algorithm	Can be mixed in the same applications, but require attention or modest modifications
Run	Fixes to bugs and minor issues. Later run automatically replaces former	Consider it as a drop-in replacement

3.1.1.3 Product Content

Each netCDF4 data file contains the following variables:

- Three coordinate variables and associated dimensions:
  - Latitude: (lat) and longitude (lon)
  - Time
- A variable “crs” describing the grid mapping (coordinate reference system)
- Eight (8) SWI, one per T-value: SWI_001 (for T=1), SWI_005 (T=5) and so on.
- Eight (8) Quality flags (QFLAG), also one per T-value: QFLAG_001, QGLAG_005 etc.
- The Surface State Flag (SSF)

The netCDF files contain a number of metadata attributes at different levels:

- globally, for the entire file (Table 3)
- at the level of the SWI, QFLAG and SSF variables, as illustrated in Table 4 for the SWI_001 variable
- for the latitude (Table 5), longitude and time (Table 6) coordinate variables
- for the grid mapping (spatial reference system) variable (Table 7).

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Table 3: Description of netCDF file attributes

Attribute name	Description	Data Type	Example
Conventions	Version of the CF-Conventions used	String	CF-1.6
archive_facility	Institution that archives the product	String	VITO
copyright	Copyright notice, to be used when referring to the data source of this product in publications	String	Copernicus Service information 2025
geospatial_lat_max	Minimum Latitude	Float	89.95
geospatial_lat_min	Maximum Latitude	Float	-89.95
geospatial_lon_max	Minimum Longitude	Float	179.95
geospatial_lon_min	Maximum Longitude	Float	-179.95
history	Describes the lineage	String	2025-06-25 -- SWI NRT Production
identifier	Unique identifier for the product	String	urn:cgls:global:swi_v4_0.1degree:SWI_202507151200_GLOBE_ASCAT_V4.0.1
institution	Institution that produced the product	String	GSA
long_name	Extended product name	String	Soil Water Index
orbit_type	Orbit type of the orbiting platform(s)	String	LEO
parent_identifier	Identifier of the product collection for the product in Copernicus Global Land Service metadata catalogue	String	urn:cgls:global:swi_v4_0.1degree
platform	Name(s) of the orbiting platform(s) used.	String	METOP A+B+C
pixel_size	Size of a single pixel (in square)	String	0.1 degrees
processing_level	Product processing level	String	L3
processing_mode	Processing mode used when generating the product. One of Near-Real Time, Consolidated, Offline or Reprocessing	String	Near-Real-Time
product_version	Version of the product (VM.m.r)	String	V4.0.1
references	Published references that describe the data or methods used to produce it.	String	https://land.copernicus.eu/global/products/swi
region_name	Name of the geographic region covered	String	GLOBE
sensor	Name(s) of the sensor(s) used	String	ASCAT
source	The method of production of the original data	String	Derived from EO microwave scatterometer observations
time_coverage_end	End date and time of the temporal coverage of the input data	String	2025-07-15T12:00:00Z
time_coverage_start	Start date and time of the temporal coverage of the input data.	String	2025-07-14T12:00:00Z
title	A description of the contents of the file	String	Daily Soil Water Index 0.1degree: GLOBE 2025-07-15T12:00:00Z

Table 4: Description of netCDF attributes for variable SWI_001 (T=1)

Attribute	Description	Data Type	Examples values
CLASS	Dataset type		DATA
_FillValue	Single value used to represent missing or undefined data and to pre-fill memory space in case a non-written part of data is read back. Value must be outside of valid_range.	Same as data variable	255
coordinates	References the coordinate variables	String	time lat lon
grid_mapping	Reference to the grid mapping variable	String	crs
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	String	Soil Water Index with T=1
missing_value	Single value used to represent missing or undefined data, for applications following older versions of the standards. Value must be outside of valid_range.	Same as data variable	255
scale_factor	Multiplication factor for the variable's contents that must be applied in order to obtain the real values. Omit in case the scale is 1.	Float	0.5
units	Units of the variable's content, taken from Unidata's udunits library. Empty or omitted for dimensionless variables. Fractions should be indicated by scale_factor.	String	%
valid_range	Smallest and largest values for the variable. Missing data is to be represented by one or several values outside of this range.	Same as data variable	0, 200

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Table 5: Description of netCDF attributes for coordinate dimensions (latitudes and longitudes)

Attribute	Description	Data Type	Example
CLASS	Dataset type	String	DIMENSION_SCALE
standard_name	A standardized name that references a description of a variable's content in CF-Convention's standard names table. Note that each standard_name has corresponding unit (from Unidata's udunits).	String	longitude
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Used when a standard name is not available (FDOB layers).	String	longitude
units	Units of a the variable's content, taken from Unidata's udunits library.	String	degrees_east
axis	Identifies latitude, longitude, vertical, or time axes.	String	X
_CoordinateAxisType	Label used in GDAL library	String	Lon
valid_range	Smallest and largest values for the variable.	Same as variable	-90, 90

Table 6: Description of netCDF attributes for time coordinate variable

Attribute	Description	Data Type	Example
CLASS	Dataset type	String	DIMENSION_SCALE
standard_name	A standardized name that references a description of a variable's content in CF-Convention's standard names table. Note that each standard_name has corresponding unit (from Unidata's udunits).	String	time
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	String	time
units	Units of a the variable's content, taken from Unidata's udunits library.	String	Days since 1970-01-01 12:00:00
axis	Identifies latitude, longitude, vertical, or time axes.	String	T
calendar	Specific calendar assigned to the time variable, taken from the Unidata's udunits library.	String	standard

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Table 7: Description of netCDF attributes for the grid mapping variable

Attribute	Description	Data Type	Example
GeoTransform	Six coefficients for the affine transformation from pixel/line space to coordinate space, as defined in GDAL's GeoTransform	String	-180 0.1 0 90 0 -0.1
grid_mapping_name	Name used to identify the grid mapping	String	latitude_longitude
inverse_flattening	Used to specify the inverse flattening (1/f) of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	Float	298.257223563
long_name	A descriptive name that indicates a variable's content.	String	coordinate reference system
longitude_of_prime_meridian	Specifies the longitude, with respect to Greenwich, of the prime meridian associated with the geodetic datum	Float	0.0
semi_major_axis	Specifies the length, in metres, of the semi-major axis of the ellipsoidal figure associated with the geodetic datum and used to approximate the shape of the Earth	Float	6378137.0
spatial_ref	Spatial reference system in OGC's Well-Known Text (WKT) format	String	GEOGCS["WGS 84",DATUM["WGS_1984",… AUTHORITY["EPSG","4326"]]

The data encoding of various variables is described in Table 8. The Physical values (PhyVal) are retrieved from the Digital Number (DN, raw pixel value in the file) as follows:

$$PhyVal = DN * Scale_factor \quad \text{(Eq. 5)}$$

This scaling only applies to raw pixel values inside the range indicated by the valid_range attribute (0…200 for SWI) (Table 4), hence not the missing data value. Minimum and maximum physical values in Table 8 are valid after applying the scaling factor.

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Table 8: Data encoding of variables contained in the SWI files

Variable	Type	Scale factor	Min physical value	Max physical value	Unit	Missing Value
SWI_001	Unsigned byte	0.5	0	100	%	255
SWI_005	Unsigned byte	0.5	0	100	%	255
SWI_010	Unsigned byte	0.5	0	100	%	255
SWI_015	Unsigned byte	0.5	0	100	%	255
SWI_020	Unsigned byte	0.5	0	100	%	255
SWI_040	Unsigned byte	0.5	0	100	%	255
SWI_060	Unsigned byte	0.5	0	100	%	255
SWI_100	Unsigned byte	0.5	0	100	%	255
QFLAG_001	Unsigned byte	0.5	0	100	%	255
QFLAG_005	Unsigned byte	0.5	0	100	%	255
QFLAG_010	Unsigned byte	0.5	0	100	%	255
QFLAG_015	Unsigned byte	0.5	0	100	%	255
QFLAG_020	Unsigned byte	0.5	0	100	%	255
QFLAG_040	Unsigned byte	0.5	0	100	%	255
QFLAG_060	Unsigned byte	0.5	0	100	%	255
QFLAG_100	Unsigned byte	0.5	0	100	%	255
SSF	Unsigned byte	1	0	4		255
lon	Double float	1	-180	180	° East	-
lat	Double float	1	-180	180	° North	-
time	Double float	1	-	-	Days since 1970-01-01 12UTC	-

3.1.2 SWI10

3.1.2.1 Overview

The Near-Real Time SWI10 product contains decadal (10-daily) synthesis of Soil Water Index. The values are given on a global 0.1° regular grid of 1800 lines and 3600 columns. For each node (pixel), the product provides eight SWI10 values, eight associated quality flags (QFLAG) and eight percentages of valid observations (VOBS), one for each characteristic time length T (1, 5, 10, 15, 20, 40, 60 and 100 days). Three files, containing the above values, are available per month.

3.1.2.2 File Format

The SWI10 products are provided in netCDF4 format. The filename respects the following convention:

c_gls_SWI10_YYYYMMDDHHMM_GLOBE_ASCAT_VX.Y.Z.nc

In all filenames the following placeholders will be replaced by actual values:

- YYYYMMDD is the date of the product, with DD fixed to days 01, 11 and 21.
- HHMM is set to 1200 to indicate that the product is derived from the daily products that are representative of 12:00 UTC
- VX.Y.Z is the version in which X is the major version, Y the minor version and Z the run number. The recommended approach for handling updates to the versioning is the same as for the daily SWI products as outlined in Table 2.

3.1.2.3 Product Content

Each netCDF4 data file contains the following variables:

- Three coordinate variables and associated dimensions:
  - Latitude (lat) and longitude (lon)
  - Time
- A variable “crs” describing the grid mapping (coordinate reference system)
- Eight (8) variables with 10-daily SWI values, one per T-value, named SWI_001 for T=1, SWI_005 for T=5 and so on.
- Eight (8) variables with Quality flags (QFLAG), one per T-value, named QFLAG_001, QFLAG_005, etc.
- Eight (8) variables with percentages of Valid Observations (VOBS), also one per T-value

The netCDF files contain a number of metadata attributes at different levels:

- globally, for the entire file (Table 9)
- at the level of the SWI10, QFLAG and VOBS variables
- at the level of the three coordinate variables (lat, lon, time)
- at the level of the grid mapping variable (crs)

The netCDF attributes for the SWI10, QFLAG and VOBS data, as well as for the latitude, longitude and time coordinate variables and the grid mapping variable are similar to those used in the daily SWI products (section 3.1.1.3, in particular Table 4, Table 5, Table 6 and Table 7).

Table 9: List of attributes of the global level in the SWI10 files

Attribute name	Description	Data Type	Example
Conventions	Version of the CF-Conventions used	String	CF-1.6
archive_facility	Institution that archives the product	String	VITO
copyright	Copyright notice, to be used when referring to the data source of this product in publications	String	Copernicus Service information 2025
geospatial_lat_max	Minimum Latitude	Float	89.95
geospatial_lat_min	Maximum Latitude	Float	-89.95
geospatial_lon_max	Minimum Longitude	Float	179.95
geospatial_lon_min	Maximum Longitude	Float	-179.95
history	Describes the lineage	String	2025-06-25 -- SWI NRT Production
identifier	Unique identifier for the product	String	urn:cgls:global:swi10_v4_0.1degree:SWI10_202507111200_GLOBE_ASCAT_V4.0.1
institution	Institution that produced the product	String	GSA
long_name	Extended product name	String	Soil Water Index
orbit_type	Orbit type of the orbiting platform(s)	String	LEO
parent_identifier	Identifier of the product collection for the product in Copernicus Global Land Service metadata catalogue	String	urn:cgls:global:swi10_v4_0.1degree
platform	Name(s) of the orbiting platform(s) used.	String	METOP A+B+C
pixel_size	Size of a single pixel (in square)	String	0.1 degrees
processing_level	Product processing level	String	L3
processing_mode	Processing mode used when generating the product. One of Near-Real Time, Consolidated, Offline or Reprocessing	String	Reprocessing
product_version	Version of the product (VM.m.r)	String	V4.0.1
references	Published references that describe the data or methods used to produce it.	String	https://land.copernicus.eu/global/products/swi
region_name	Name of the geographic region covered	String	GLOBE
sensor	Name(s) of the sensor(s) used	String	ASCAT
source	The method of production of the original data	String	Derived from EO microwave scatterometer observations
time_coverage_end	End date and time of the temporal coverage of the input data	String	2025-07-20T12:00:00Z
time_coverage_start	Start date and time of the temporal coverage of the input data.	String	2025-07-10T12:00:00Z
title	A description of the contents of the file	String	10-daily Soil Water Index 0.1degree: GLOBE 2025-07-11T12:00:00Z

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The data encoding of the variables is described in Table 10. The Physical values (PhyVal) are retrieved from the Digital Number (DN, raw pixel value in the file) as (Eq.6). The scaling only applies to raw pixel values inside the range indicated by the valid_range attribute (0…200 for SWI) (Table 4), hence not the missing data value. Minimum and maximum physical values in Table 10 are valid after applying the scaling.

Table 10: Data encoding of variables contained in the SWI10 files

Variable	Type	Scale factor	Min physical value	Max physical value	Unit	Missing Value
SWI_001	Unsigned byte	0.5	0	100	%	255
SWI_005	Unsigned byte	0.5	0	100	%	255
SWI_010	Unsigned byte	0.5	0	100	%	255
SWI_015	Unsigned byte	0.5	0	100	%	255
SWI_020	Unsigned byte	0.5	0	100	%	255
SWI_040	Unsigned byte	0.5	0	100	%	255
SWI_060	Unsigned byte	0.5	0	100	%	255
SWI_100	Unsigned byte	0.5	0	100	%	255
QFLAG_001	Unsigned byte	0.5	0	100	%	255
QFLAG_005	Unsigned byte	0.5	0	100	%	255
QFLAG_010	Unsigned byte	0.5	0	100	%	255
QFLAG_015	Unsigned byte	0.5	0	100	%	255
QFLAG_020	Unsigned byte	0.5	0	100	%	255
QFLAG_040	Unsigned byte	0.5	0	100	%	255
QFLAG_060	Unsigned byte	0.5	0	100	%	255
QFLAG_100	Unsigned byte	0.5	0	100	%	255
VOBS_001	Unsigned byte	1	0	100	%	255
VOBS_005	Unsigned byte	1	0	100	%	255
VOBS_010	Unsigned byte	1	0	100	%	255
VOBS_015	Unsigned byte	1	0	100	%	255
VOBS_020	Unsigned byte	1	0	100	%	255
VOBS_040	Unsigned byte	1	0	100	%	255
VOBS_060	Unsigned byte	1	0	100	%	255
VOBS_100	Unsigned byte	1	0	100	%	255
lon	Double float	1	-180	180	° East	-
lat	Double float	1	-180	180	° North	-
time	Double float	1	-	-	Days since 1970-01-01 12UTC	-

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3.1.3 SWI Static Layers dataset

3.1.3.1 File Format

The ancillary dataset, described in Section 5.1.2, can be found in the SWI Static Layers dataset on the CLMS web portal[4]. They are provided as NetCDF files. Their filenames follow the following convention:

c_gls_SWI-STATIC-XX_YYYYMMDDHHMM_GLOBE_SENSOR_VX.Y.Z

- YYYYMMDDHHMM is the date and time of the product:
  - The fixed date 20150101 for the Tropical Forest Mask, Topographic Complexity and Wetland Fraction
  - The start of the year (e.g. 200701010000) for the annual Correlation Information (CI).
  - The range of complete years used (e.g. 2007----2017) for the multi-year CI
- VX.Y.Z is the version in which X is the major version, Y the minor version and Z the run number. The recommended approach for handling updates to the versioning is the same as for the daily SWI products as outlined in Table 2.

Table 11 contains the possible field combinations of XX and SENSOR.

Table 11: File naming of STATIC dataset

Dataset/Variable name	XX	SENSOR
Topographic Complexity	TC	GTOPO30
Wetland Fraction	WF	GLWD
Tropical Forest Mask	FM	ASCAT
Correlation Information	CI	SWI-GLDAS

3.1.3.2 Product Content

Each netCDF4 data file contains the following variables:

- Coordinate variables for latitude (lat) and longitude (lon)
- A grid mapping variable (crs)
- A main data variable, named SWI_STATIC_XX with XX from Table 11.

As these products are static and not designed to be stacked over time to form a time series, they do not contain a time dimension and coordinate variable.

[4] https://land.copernicus.eu/en/products/soil-moisture/soil-water-index-static-layers-12-5-km

The netCDF files contain metadata attributes at the global file level as well as the level of each of the variables. Table 12 provides an example of the global attributes for the Tropical Forest Mask product. The Correlation Information adds two attributes – *time_coverage_start* and *time_coverage_end* to describe the time period the CI is computed for. Specific attributes, notably *sensor* and *platform* can be set to the value *N/A* when they do not apply.

The netCDF attributes for the latitude and longitude coordinate variables and the grid mapping variable are similar to those used in the daily SWI products (section 3.1.1.3, in particular Table 4, Table 5 and Table 7).

Table 12: List of attributes of the global level in the Tropical Forest Mask product

Attribute name	Description	Data Type	Example
Conventions	Version of the CF-Conventions used	String	CF-1.6
archive_facility	Institution that archives the product	String	VITO
copyright	Copyright notice, to be used when referring to the data source of this product in publications	String	Copernicus Service information 2018
geospatial_lat_max	Minimum Latitude	Float	89.95
geospatial_lat_min	Maximum Latitude	Float	-89.95
geospatial_lon_max	Minimum Longitude	Float	179.95
geospatial_lon_min	Maximum Longitude	Float	-179.95
history	Describes the lineage	String	2018-09-01 -- Reformatted to netCDF
identifier	Unique identifier for the product	String	urn:cgls:global:swi-static-fm_v3_0.1degree:SWI-STATIC-FM_201501010000_GLOBE_ASCAT_V3.2.1
institution	Institution that produced the product	String	TU Wien
long_name	Extended product name	String	Soil Water Index
orbit_type	Orbit type of the orbiting platform(s)	String	LEO
parent_identifier	Identifier of the product collection for the product in Copernicus Global Land Service metadata catalogue	String	urn:cgls:global:swi-static-fm_v3_0.1degree
platform	Name(s) of the orbiting platform(s) used. One of “Metop A”, “Metop B” and “Metop A+B”	String	METOP A
pixel_size	Size of a single pixel (in square)	String	0.1 degrees
processing_mode	Processing mode used when generating the product. One of Near-Real Time, Consolidated, Offline or Reprocessing	String	Offline
product_version	Version of the product (VM.m.r)	String	V3.2.1

Attribute name	Description	Data Type	Example
references	Published references that describe the data or methods used to produce it.	String	https://land.copernicus.eu/en/products/soil-moisture/soil-water-index-static-layers-v3-0-12-5-km
region_name	Name of the geographic region covered	String	GLOBE
sensor	Name(s) of the sensor(s) used	String	ASCAT
source	The method of production of the original data	String	Derived from EO microwave scatterometer observations
title	A description of the contents of the file	String	SWI Static layer: ancillary Tropical Forest Mask

The data encoding of variables is described in Table 13. The values can be read without applying any scaling.

Table 13: Data encoding of the main data variables in the SWI-Static files

Product / Variable	Type	Min physical value	Max physical value	Unit	Missing Value
Tropical Forest Mask / SWI_STATIC_FM	Unsigned byte	0	1		255
Wetland Fraction / SWI_STATIC_WF	Unsigned byte	0	100	%	255
Topographic Complexity / SWI_STATIC_TC	Unsigned byte	0	100	%	255
Correlation Information / Correlation_SWI_GLDAS	Double Float	-1	1		NaN

3.1.4 Viewing the data

3.1.4.1 Viewing in QGIS

Up to date version of QGIS[5] supports viewing of netCDF4 data. Just open the .nc file with QGIS and select a layer to view a selected variable.

3.1.4.2 Viewing in Panoply

NetCDF4 files can be easily displayed as map or as time series with the Panoply-tool[6]. For instance, the plots displayed in Section 5.1.2 are generated with this software.

[5] https://www.qgis.org/en/site/
[6] http://www.giss.nasa.gov/tools/panoply/

3.1.5 Product Characteristics

3.1.5.1 Projection and grid information

The product is displayed in a regular latitude/longitude grid (plate carrée) with the ellipsoid WGS 1984 (Terrestrial radius=6378km).

The resolution of the grid is 0.1°.

As the data covers the whole globe, datasets are arrays of 1800 lines per 3600 columns.

3.1.5.2 Temporal information

The SWI product is a daily composite made using the SSM acquired between Day1 at 12:00 UTC and Day2 at 11:59 UTC. The part *"YYYYMMDDHHMM"* of the product name gives the date of Day2. The SWI products are updated every day.

The SWI10 product is a dekadal (10-daily) composite which is made using the SWI products of days 1 to 10, days 11 to 20, days 21 to end of month for each month of the year. The part *"YYYYMMDDHHMM"* of the product name gives the date of the first day of the dekad.

The daily SWI and 10-daily SWI10 products are provided for the period from January 2007 to present.

The Tropical Forest Mask, Topographic Complexity and Wetland Fraction products are static and can be used in the analysis of any SWI or SWI10 product.

The Correlation Information product provides information for either one year, with the start date of that year given in the filename, or for a set of full years (indicated as Year1----Year2 in the filename).

3.2 TIME SERIES PRODUCTS

3.2.1 SWI Time Series (SWI-TS)

3.2.1.1 Overview

The SWI-TS product is a reformatted SWI image product which makes time series analysis on the local or global scale easier. It is saved in netCDF Format according to the netCDF Climate and Forecast (CF) Metadata Convention (CF conventions). The CF conventions define multiple possible representations of time series data. Since the SWI data is equally spaced and all time series are of equal length the "Orthogonal multidimensional array representation" was chosen (see chapter 9.3.1 of Eaton et al. 2011).

3.2.1.2 File Format

The time series are provided as netCDF data files for each of the cells in Figure 6. Each cell file has the filename:

c_gls_SWI-TS_YYYYMMDDHHMM_CDDDD_ASCAT_VX.Y.Z.nc

where

- YYYYMMDDHHMM specifies the date of the last observation in the SWI-TS file
- DDDD is the cell number
- VX.Y.Z is the version in which X is the major version, Y the minor version and Z the run number. The recommended approach for handling updates to the versioning is the same as for the daily SWI products as outlined in Table 2.

The complete grid information as well as the cell number for each grid point can be found in the SWI Static Layers collection (Section 3.1.3). This file is useful for e.g. nearest neighbour searching across cell boundaries. This file is named:

c_gls_SWI-STATIC-DGG_YYYYMMDDHHMM_GLOBE_ASCAT_VX.Y.Z.nc

where

- YYYYMMDDHHMM specifies the date of the grid generation
- VX.Y.Z is the version in which X is the major, Y the minor and Z the run version. X and Y only change with algorithm changes whereas Z changes with processing runs

The DGG product also contains a main netCDF data file, GeoTIFF quicklook image and XML metadata file.

3.2.1.3 Product Content

Each cell file contains 2 dimensions (Table 14) and 23 Variables (Table 15). The data encoding is equal to the one of the SWI image product listed in Table 8 and documented in the netCDF metadata.

Table 14: netCDF dimensions in a cell file

Name	Size	Comment
locations	Cell dependent	Number of grid points in the cell file
time	UNLIMITED	Number of observations for each time series. Set to UNLIMITED so that the time series can be extended.

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Table 15: netCDF variables in a cell file

Name	Dimensions	Units	Type	Comment
lon	locations	degrees east	double	longitude of each location
lat	locations	degrees north	double	latitude of each location
locations	locations	N/A	long	location or grid point id
location_description	locations	N/A	string	location description, not filled in this case
time	time	day	double	Gives the observation time in days since 1900-01-01 00:00:00 for each observation.
SWI_001	locations, time	%	unsigned byte	SWI with T=1
SWI_005	locations, time	%	unsigned byte	SWI with T=5
SWI_010	locations, time	%	unsigned byte	SWI with T=10
SWI_015	locations, time	%	unsigned byte	SWI with T=15
SWI_020	locations, time	%	unsigned byte	SWI with T=20
SWI_040	locations, time	%	unsigned byte	SWI with T=40
SWI_060	locations, time	%	unsigned byte	SWI with T=60
SWI_100	locations, time	%	unsigned byte	SWI with T=100
QFLAG_001	locations, time	%	unsigned byte	QFLAG with T=1
QFLAG _005	locations, time	%	unsigned byte	QFLAG with T=5
QFLAG _010	locations, time	%	unsigned byte	QFLAG with T=10
QFLAG _015	locations, time	%	unsigned byte	QFLAG with T=15
QFLAG _020	locations, time	%	unsigned byte	QFLAG with T=20
QFLAG _040	locations, time	%	unsigned byte	QFLAG with T=40
QFLAG _060	locations, time	%	unsigned byte	QFLAG with T=60
QFLAG _100	locations, time	%	unsigned byte	QFLAG with T=100
SSF	locations, time	N/A	unsigned byte	Surface State Flag

The grid information file (DGG) contains 1 dimension (Table 16) and 5 variables (Table 17).

Table 16: netCDF dimension in SWI-STATIC_DGG file

Name	Size	Comment
locations	3264391	Number of all grid points.

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Table 17: netCDF variables in SWI-STATIC-DGG file

Name	Dimensions	Units	Type	Comment
lon	locations	degrees east	float	longitude of each location
lat	locations	degrees north	float	latitude of each location
cell	locations	N/A	short	cell number for each location
location_id	locations	N/A	long	location or grid point id
land_flag	locations	N/A	byte	Indicates if location is over land or not. Set to 0 if over water and 1 if over land.

3.2.2 Handling Time Series Data

Unfortunately, there is currently no software with a GUI that supports reading time series data stored according to the CF conventions, so the following sections explain the general approach and also point to an open-source package that can read the SWI-TS data.

The first step is finding the location id of the grid point (GP) you are interested in. Assuming you know the latitude and longitude of the GP you want to read, you can use the DGG grid point locator hosted at https://dgg.geo.tuwien.ac.at/ to find and export all location id's of a country or region. Figure 4 shows the interface. Note that location id's are called Point ID's in the viewer. The DGG grid is called "Copernicus SWI TS" in the web application. After you have found the id and the cell number of the GP you are interested in, open the corresponding cell file with the programming language of your choice. Figure 5 shows an example of reading and plotting a SWI T=20 time series in the Python programming language.

There is also the ascat python package that includes easy to use readers for various soil moisture datasets derived from the ASCAT sensor and also handles nearest neighbour search. Please consult the ascat website[7] for the latest instructions on how to use and install the package.

[7] https://github.com/TUW-GEO/ascat

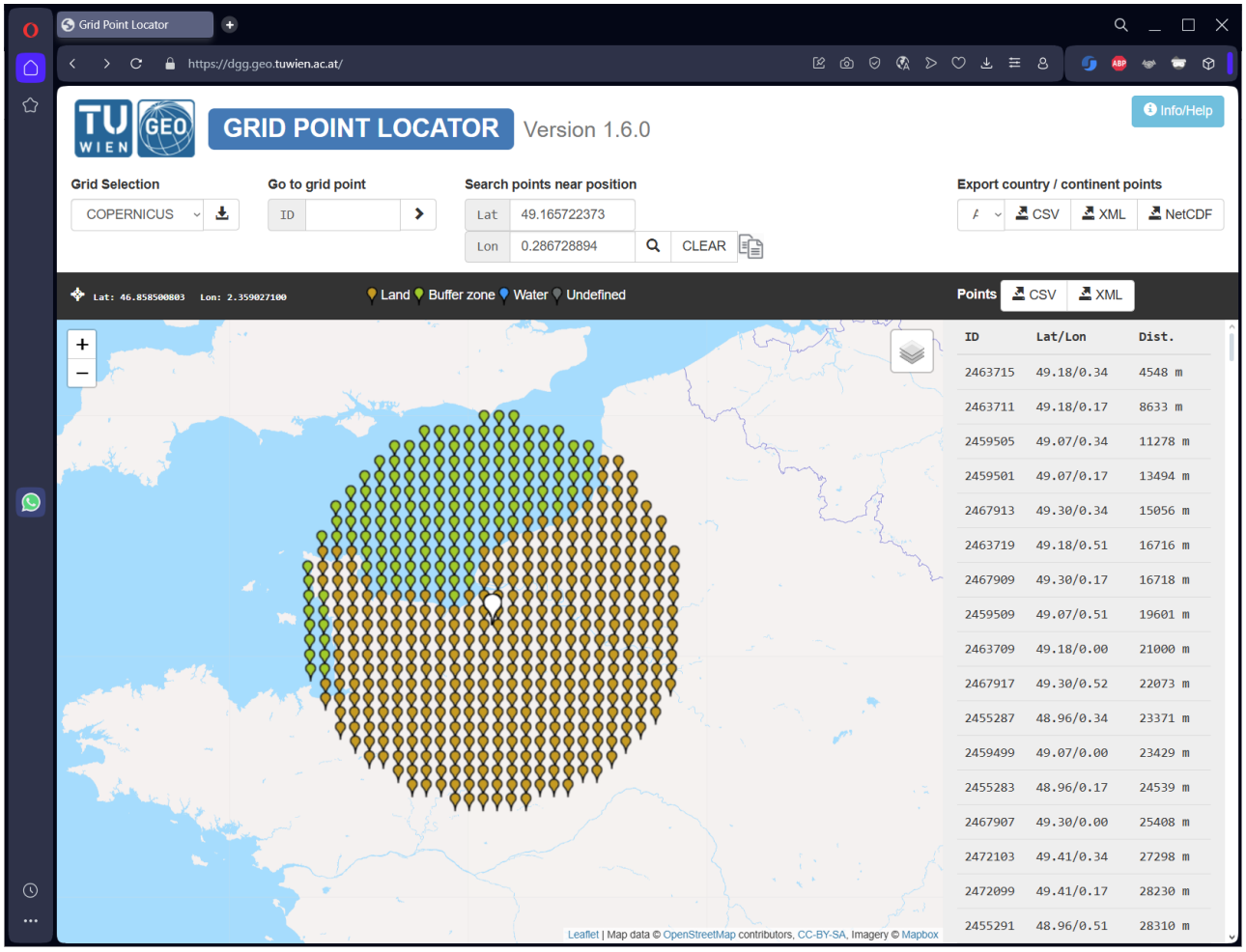


Figure 4: Screenshot of the DGG point locator.

```
import netCDF4
import os
import numpy as np
import matplotlib.pyplot as plt

cell = 355
gp = 2412919

path_to_cell_file = os.path.join(
    'path', 'to', 'SWI_TS',
    'c_gls_SWI-TS_201412310000_C%04d_ASCAT_V3.0.1.nc' % cell)

with netCDF4.Dataset(path_to_cell_file) as ncdata:
    # find the index of the gp index in the cell file
    loc_idx = np.where(ncdata.variables['locations'][:] == gp)[0]

    # each variable is a 2D array of size locations x time
    # we want the slice of the array for our loc_idx for the whole time
    # the returned arrays are 2D, we select the first dimension with [0]
```

```
swi20 = ncdata.variables['SWI_020'][loc_idx, :][0]
qflag20 = ncdata.variables['QFLAG_020'][loc_idx, :][0]
ssf = ncdata.variables['SSF'][loc_idx, :][0]
# get the time units and convert from numbers to datetime format
time_units = ncdata.variables['time'].units
time = netCDF4.num2date(ncdata.variables['time'][:],
                        time_units)
# mask SWI and QFLAG for SSF values other than 1 (unfrozen)
mask = ssf != 1
swi20[mask] = np.nan
qflag20[mask] = np.nan
# plot the time series
plt.plot(time, swi20, label='SWI T=20')
plt.plot(time, qflag20, label='QFLAG T=20')
plt.legend()
plt.show()
```

Figure 5: Python example code for reading a time series.

3.2.3 Product Characteristics

3.2.3.1 Projection and grid information

The time series are saved on a DGG with a total of 3 264 391 grid points of which 839 826 are over land. They are equally spaced with a grid spacing of 12.5km. The whole grid is partitioned in 10x10° cells (Figure 6) which means that there are 308 cells containing valid data over land.

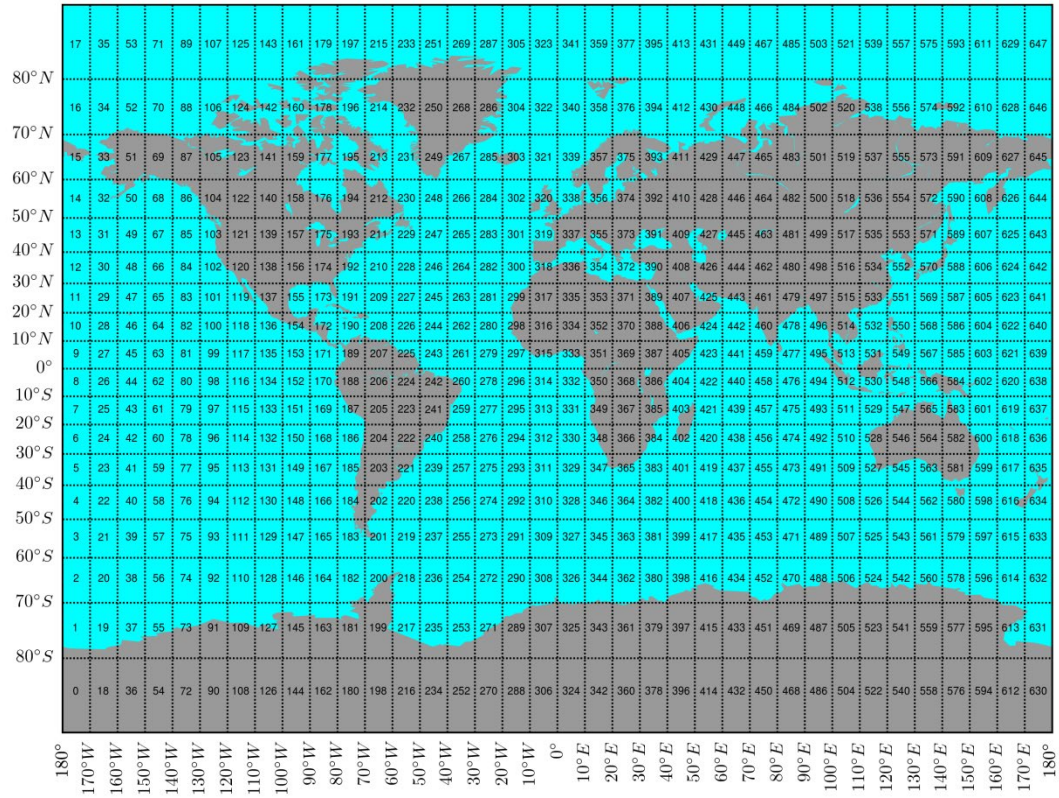


Figure 6: Cell partitioning of the DGG into 10x10° cells.

3.2.3.2 Temporal information

The SWI-TS product contains time series with a temporal sampling of 1 day. The time part of the date time stamp of each data point is 12:00 UTC. Each data point contains information from SWI data of the previous 24 hours.

The SWI-TS product is updated every six months, in January and July.

3.3 DATA POLICIES

The Copernicus component is governed by Regulation (EU) No 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme.

The Copernicus land monitoring products and services are made available on a principle of **full, open and free access**, as established by the Commission Delegated Regulation (EU) No 1159/2013 of 12 July 2013.

How to cite and reference the products and services of the Copernicus Land Monitoring Service is explained here: https://land.copernicus.eu/en/data-policy.

3.4 ACCESS AND CONTACTS

The global SWI 0.1°, SWI10 and SWI-TS products are available through the Copernicus Land Monitoring Service website at the address: https://land.copernicus.eu/ with contact information (helpdesk) on https://land.copernicus.eu/en/contact-service-helpdesk.

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4 QUALITY ASSESSMENT

4.1 OVERVIEW

The accuracy assessment of remotely sensed soil moisture is an active research topic in which no consensus is reached on how to best validate or compare different datasets (Wagner et al. 2014). Whereas some researchers use the RMSD as their leading metric (NASA 2014), others focus on the correlation coefficient (R) (H-SAF 2012). Triple collocation (TC) (Dorigo et al. 2010; Scipal et al. 2008) is also becoming commonly used as a tool for validation of remotely sensed soil moisture but errors based on TC are currently not used as user requirements. Often user requirements give a fixed RMSD for large areas, or even globally, which is not sensible (Draper et al. 2013) since the RMSD depends strongly on the dynamic range. This threshold might be easily satisfied in arid regions, while higher RMSD values would still be acceptable in areas with high soil moisture variability.

The product accuracy essentially relies on the accuracy of the input ASCAT level-2 surface soil moisture. An elaborated error analysis of this input has been performed before. It involves a Gaussian error propagation model which tracks the different sources of errors and signal noise from radar backscatter to the final surface soil moisture product (Naeimi 2009). The product quality of the updated H29 ASCAT SSM input stream has been analysed in its Product Validation Report [ASCAT_SSM_NRT_12_5km_H29_PVR_v0_1].

In an exhaustive validation exercise, the SWI V3 data was compared to in-situ and simulated soil moisture data profile. The procedure and the outcomes are reported in the Quality Assessment report [GIOGL1_QAR_SWIV3-SWI10]. An analogous exercise was carried out after SWI V4.0 became available in NRT on 14th July 2025, followed by a reprocessing of the entire SWI collection. SWI V3 fields were also directly contrasted with those of SWI V4. Procedure and outcomes are reported in the validation report [CGLOPS1_VR_SWI0.1-V4]; the following section provides a summary of the results. In addition, regular quality monitoring is performed on a yearly basis to check that the product quality remains stable over time.

4.2 VALIDATION RESULTS SUMMARY

The first validation of the SWI V4 products was performed for the long-term archive from 2007 to 2024. The SWI data were compared to the output of the Global Land Data Assimilation System (GLDAS) land surface model and to in-situ data of the International Soil Moisture network (ISMN).

Analysing differences in the obtained performance metrics against modelled and in-situ soil moisture, an overall trend towards improved performance metrics was found for the SWI V4 dataset, highlighting the effectiveness of the implemented improvements in the new surface soil moisture input data stream (HSAF H29). Particularly, the frozen surface conditions provided by the new H

SAF H29 SSM product (instead of the previously used SSF) to mask potentially irritated SSM observations have a drastic effect in regions with pronounced cold seasons. Note the substantial drying of the northern latitudes in SWI V4 in Figure 7, which compares SWI V3 against SWI V4; Figure 8 confirms that this is a change for the better, as the unbiased root-mean-square-difference (ubRMSD), with respect to GLDAS as reference, decreases there. Conversely, the other augmentations introduced in H29 SSM result in increased SWI values in wide areas around the Red Sea and the southern United States, for example.

A more comprehensive picture can be obtained by viewing time series: Figure 9 contrasts SWI V3 and SWI V4, this time with ISMN measurements at Cochora Ranch, 35.1165°N/119.572°W, California, USA as reference. By and large, the three compared data sets capture the seasonal fluctuations similarly. The correlation coefficients confirm this impression: correlation between SWI V3 and ISMN is 0.80 and R=0.77 for SWI V4 vs. ISMN. Still, there is an obvious systematic difference in dry seasons where the SWI V3 tends to go distinctly lower, close to the zero-line. As noted just before, during the discussion of Figure 7, a general moistening of the Southern United States was observed with the introduction of SWI V4. This is in agreement with the time series presented here, but its novel aspect is that not all seasons equally contribute to this net effect. Such patterns are to be understood as local: in other parts of the world, series were found where SWI V4 showed a tendency to be lower than SWI V3 in the dry season.

The annual validation of the SWI V4 product for 2025 shows a stable good performance against both model data and in-situ data. Expected weak performances are observed in subarctic regions, densely vegetated areas (tropics), deserts, and areas that show high topographic complexity.

In quantitative terms on a global view, generally more than 90% of the grid points fulfil the threshold requirement for SWI V4 when compared to GLDAS whereas the goal is fulfilled by 42-61% (period 2007-2024) or 31-37% (for 2025), respectively. Lower agreement was observed in the comparison to in-situ data: while figures are still above 90% with respect to the threshold requirement, the goal is reached by 16-34% (period 2007-2024) and 28-45% for 2025.

A stability analysis was carried out on a time series of the global bias between ISMN and SWI V4 (T=1). The null hypothesis of “no trend in the data” could not be rejected, confirming that ASCAT is a stable instrument.

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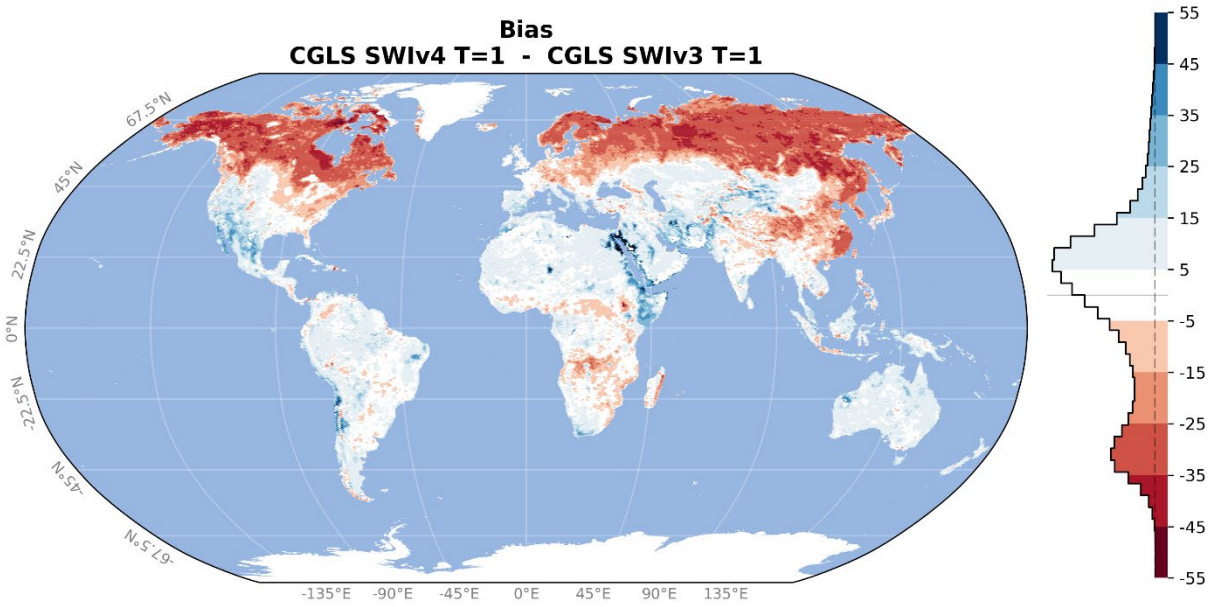


Figure 7: Bias SWI V4 T=1 minus SWI V3 T=1 for the comparison validation period 2007-2024.

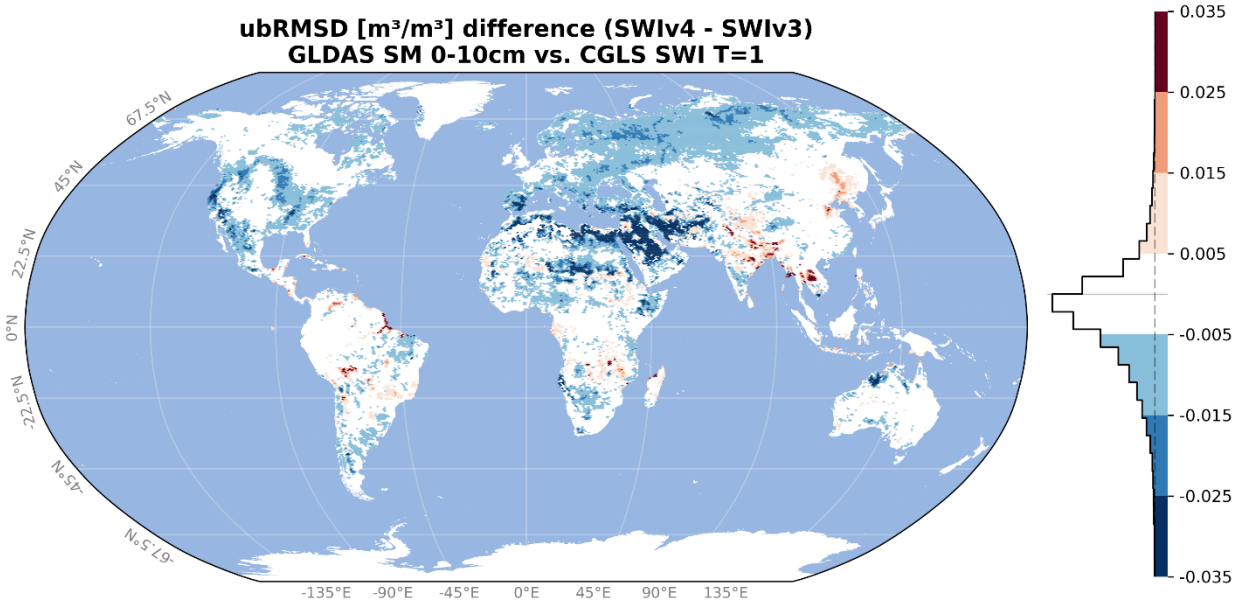


Figure 8: Difference of ubRMSD [m³/m³] (SWI V4 – SWI V3) for T=1 vs. GLDAS soil moisture 0-0.1m for the comparison validation period (2007-2024)

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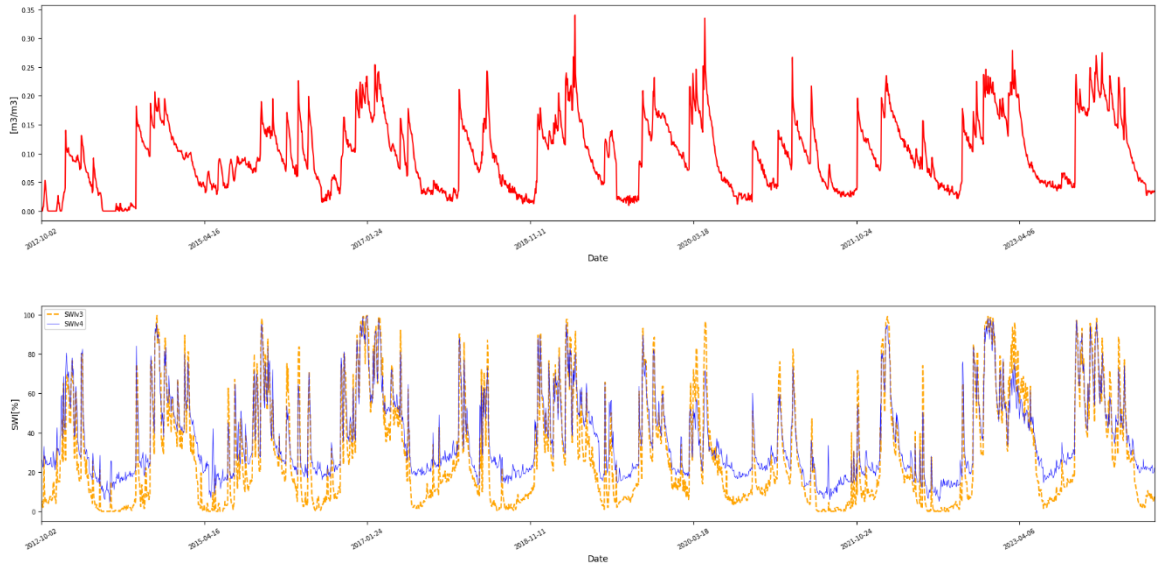


Figure 9: Upper panel: Soil moisture time series (in m^3/m^3) at the ISMN station Cochora Ranch, California. Lower panel: Corresponding time series for SWI V3 (orange) and SWI V4 (blue), in % saturation, both with T=1.

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5 PRODUCT USAGE

This chapter brings together practical guidance for using the SWI well. Section 5.1 covers how to select a T-value and how to recognise the product's limitations, applying the masking and quality layers where the SWI is less reliable. Section 5.2 describes the sectors and applications the product is suited to. How to read the files, including scaling and fill values, is described in Chapter 3.

5.1 RECOMMENDATIONS ON USING THE PRODUCT

5.1.1 Selecting the T-value

Giving a general rule on how to translate a given T-value to a certain soil depth is currently not possible since this depends strongly on the application and the soil composition of the area of interest. (Paulik et al. 2014) looked at which T-value correlates best to in situ observations of different observation depths up to 0.5m. They found that for 80.54% of the time series tested a T-value of greater than 1 showed the best correlation. They also found a weak (R^2=0.115, p-value=0.028) linear relationship between in situ measurement depth and T-value of

$$\boldsymbol{T = -56.68 * d + 11.04} \quad \text{(Eq. 6)}$$

where *d is* the in-situ observation depth in meters.

Table 18: Excerpt of Table I of (de Lange et al. 2008) showing optimized T-values per soil texture class and sampling interval.

Sampling Interval (day)	Sand	Loamy sand	Sandy loam	Loam	Silt loam	Silt	Sandy clay loam	Clay loam	Silty clay loam	Sandy clay	Silty clay	Clay
1	7	9	11	11	9	10	10	7	5	9	2	2
2	8	10	12	11	10	10	11	8	6	9	3	3

De Lange et al. (2008) found that the calibration of the T-value depends not only on soil texture but also on the sampling interval of the data. For a sampling of approx. 5 days a T-value of 20 was found to work well in most situations. For shorter sampling intervals a calibration based on soil textures should provide a benefit based on the residual sum of squares that was used in (de Lange et al. 2008). Table 18 shows an excerpt of the presented optimized T-values per soil texture class and sampling interval.

Ceballos et al. (2005) found that for their study region in Spain the best T-values were 40, 50 and 60 for layer depths of 0-25cm, 0-100cm and 50-100cm respectively.

5.1.2 Masking and quality filtering

The SWI is not equally reliable everywhere, so we recommend masking out the affected areas. The SSF gives information on frozen and snow-covered surface, and is used only for masking SSM data in SWI calculations in products prior to version 4.0. It cannot always distinguish between frozen soil and standing surface water, so the latter might be masked incorrectly. The product version 4.0 uses the H29 ASCAT SSM advisory flags directly to mask SSM for frozen soil conditions when calculating the SWI. Masking the SWI itself is left to the users. SWI data can be masked by using the quality flag (Section 2.2.1), SSF data (Section 2.2.2) or the ancillary datasets which are delivered together with the SWI or by using external datasets. Four static ancillary data are delivered together with the SWI and their retrieval and usage are explained in the next sections.

5.1.2.1 Topographic complexity

Soil moisture retrieval in mountainous regions can be difficult because of various error sources on the radar signal. Calibration errors due to the deviation from the reference ellipsoid influence of snow and ice cover as well as reduced sensitivity due to forest and rock are the main error sources.

The topographic complexity was calculated from GTOPO30 (USGS 1996) data which is a global digital elevation model with 30 arc seconds resolution (approx. 1km). For each grid point the standard deviation of elevation was calculated. The results were then normalized to values from 0 to 100 (Scipal et al. 2005). Figure 10 shows the flag on a map.

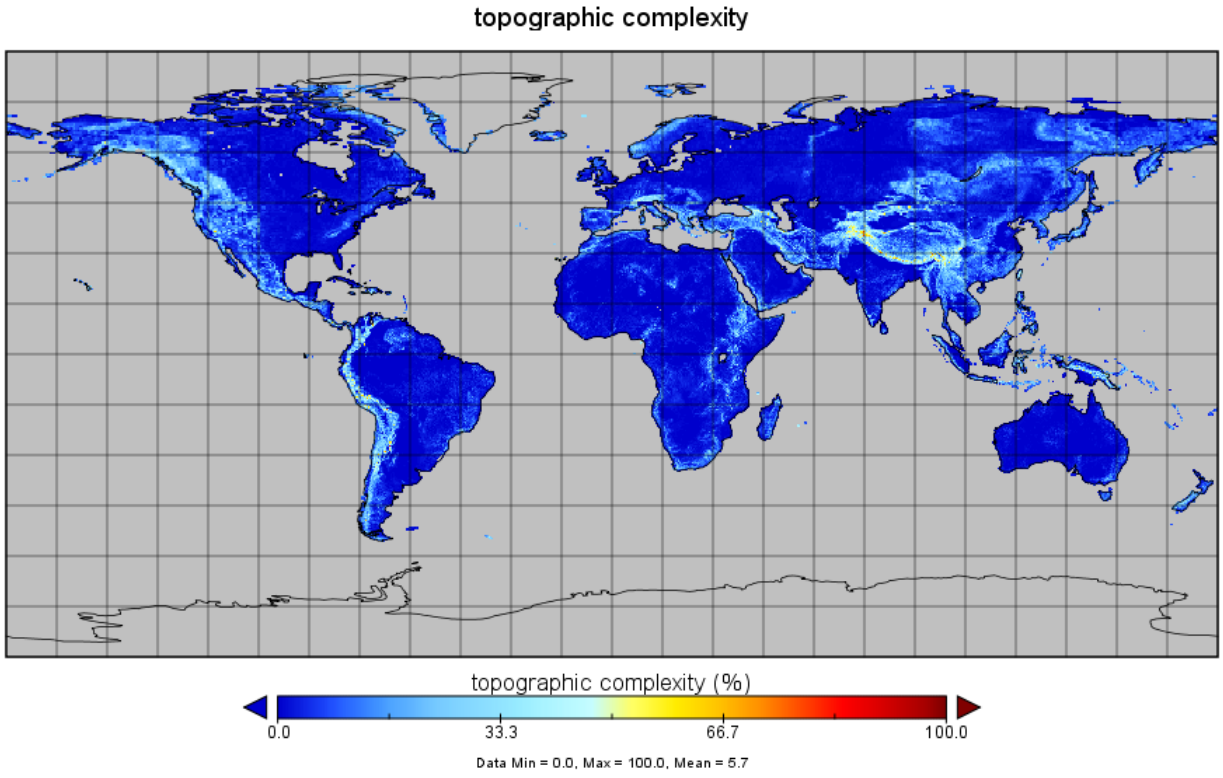


Figure 10: Topographic complexity in percent.

5.1.2.2 Wetland fraction

The wetland fraction is based on the Global Lake and Wetlands Database (GLWD) (Lehner and Döll 2004). It gives the fraction of each grid cell that is covered by water in percent from 0 to 100. Figure 11 shows the wetland fraction on a map.

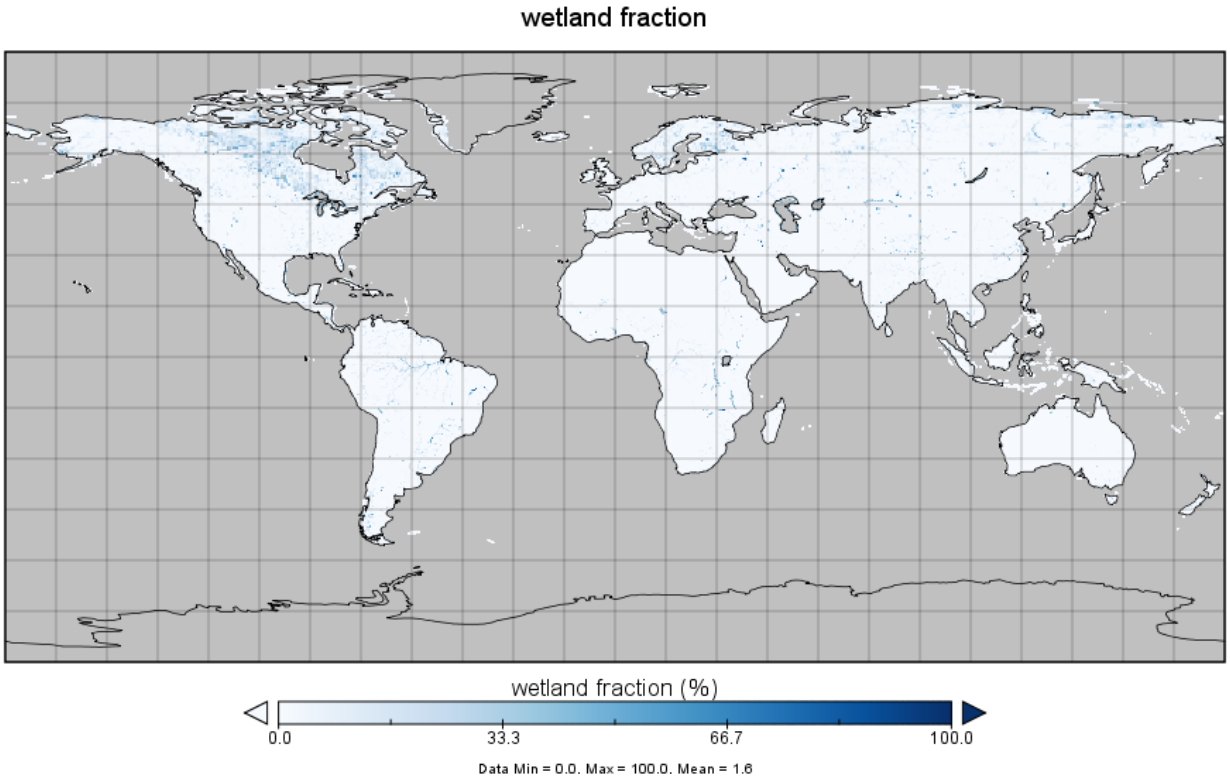


Figure 11: Wetland fraction in percent.

5.1.2.3 Tropical forest mask

The tropical forest mask is based upon the fact that radar backscatter from C-Band is mostly stable over tropical rain forests (Birrer et al. 1982; Crapolicchio and Lecomte 2003). For every grid point, it is checked if the wet and dry references of the SSM algorithm are below 31.9dB and if the sensitivity (difference between highest and lowest backscatter observation) is below 1dB. If these thresholds are met, it is assumed that the grid point is located over tropical rainforest. Figure 12 shows the tropical forest mask on a map.

There are also forests other than tropical forests which might hamper soil moisture retrieval to a certain degree. If it is not clear if this is the case in your area of interest, consult an independently derived forest cover map like e.g. (Hansen et al. 2013).

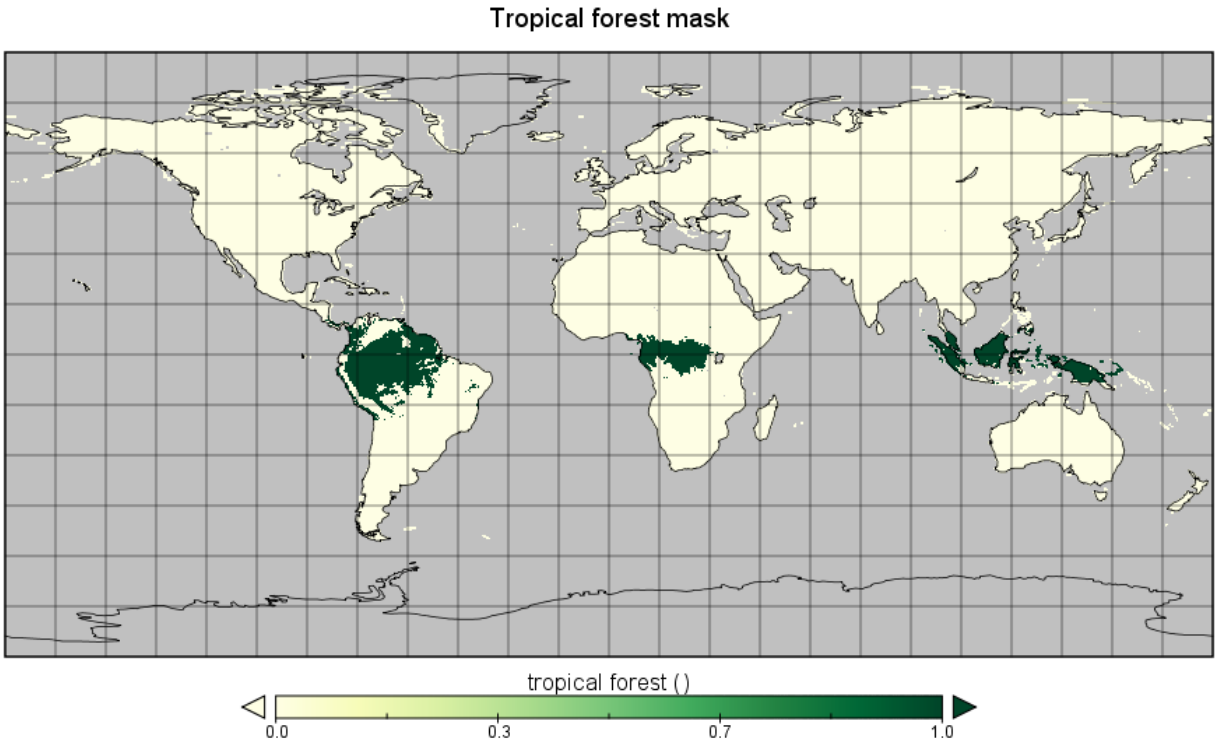


Figure 12: Tropical forest mask.

5.1.2.4 Correlation information

The quality of the SWI products varies between climate zones and biomes, as the interaction of microwaves with vegetation and with frozen and very dry soils is not straightforward and not always correctly modelled by the algorithm. To give an indication of the quality, the Correlation Information (CI) is provided to the users. These layers describe the Pearson correlation coefficient (R) of the SWI-TS product at T=1 with the upper layer soil moisture from the GLDAS (Global Land Data Assimilation System) Noah Land Surface Model (Rodell et al., 2004).

The Pearson correlation coefficient map quantifies, per pixel, the statistical agreement of the SWI products with the model data and, thus, indicates areas where the SWI is reliable. The CI aims to give guidance to the users, which need to decide, depending on the application, which R-value is acceptable, while bearing in mind that the GLDAS model has its caveats, too. The CI layers are an output of the ongoing quality assessment exercises, as presented in Chapter 4.

There is a CI layer for the period 2007-2017, describing the correlation between GLDAS and SWI-TS at T=1 over this full period (see Figure 13), and there are CI layers for each individual year. The latter can give guidance on the performance of the particular year, which may be especially important in sub-arctic areas.

Pixels without valid data from either SWI-TS, GLDAS, or where the correlation is insignificant at the level p=0.05, are set to NaN in the CI layers.

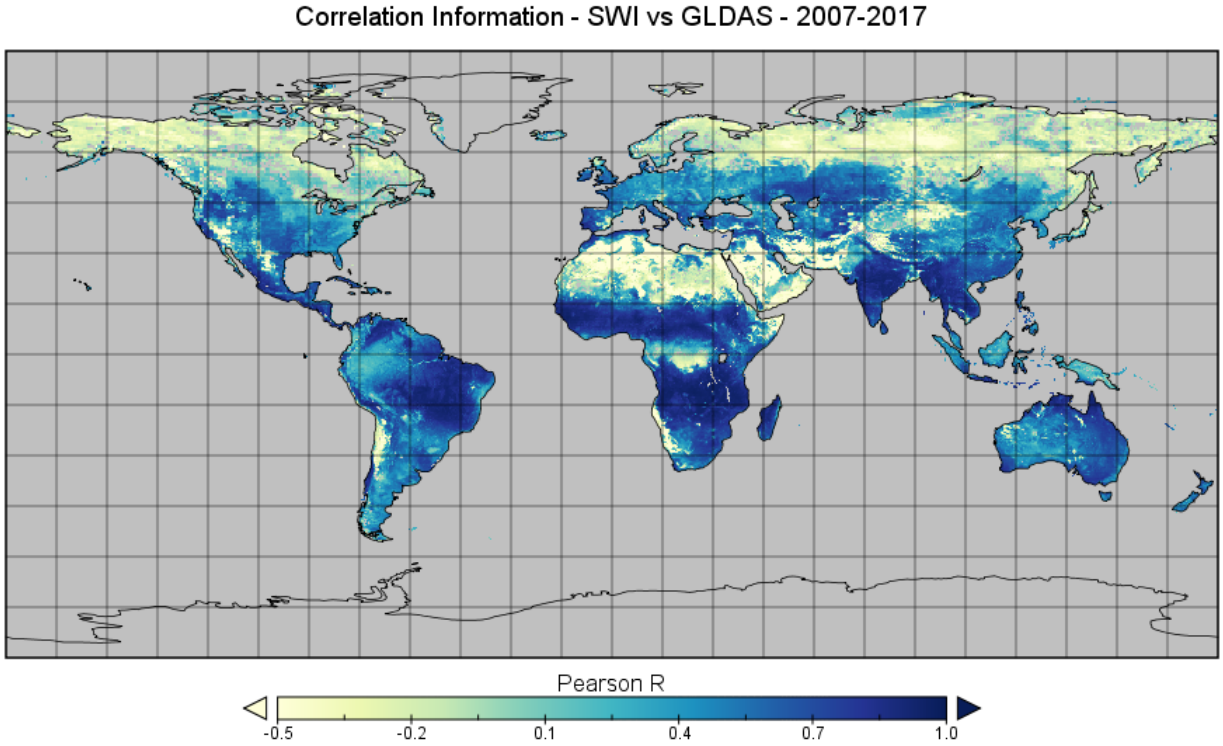


Figure 13: Correlation Information layer for the period 2007-2017.

The four layers described above flag where the SWI may be unreliable, however, this leaves the question of which cut-off values to apply. It is not straightforward to give strict thresholds of when to use the SWI depending on the ancillary data since this depends strongly on the application. (Paulik et al. 2014) found that, when comparing SWI to in situ observations, using thresholds of 20 and 15 percent, for topographic complexity and wetland fraction respectively, improved Pearson's correlation for SWI (SWI anomalies) from 0.54 (0.38) to 0.57 (0.41). Average RMSD was reduced from 0.062 to 0.056 m^3/m^3. In this study also mean soil moisture noise of above 30% was masked, which is not available as static layer.

5.1.3 Conditions to be aware of

The SWI is not reliable under all conditions. It cannot be retrieved where ASCAT surface soil moisture cannot be derived, including deserts, dense forest such as tropical rainforest, permafrost, glaciers and ice caps. It is also less reliable where processes the method does not model dominate the soil-water balance, such as strong evapotranspiration or a shallow groundwater table. Section 2.3 describes these limitations in detail, and Section 5.1.2 covers the layers for masking the affected grid cells.

5.2 SUITABILITY FOR SPECIFIC SECTORS AND APPLICATIONS

The SWI is useful in any sector where the amount of water in the soil profile matters, for example in agriculture, hydrology and natural hazards. Soil moisture is an Essential Climate Variable (GCOS), and the SWI can estimate it down to the root zone under the right conditions and within the limitations of the product (Section 2.3). The eight T-values represent different characteristic time lengths, letting users choose the one that suits their application (Section 5.1.1).

Root-zone soil moisture measures the water available to plants, which makes the SWI useful in agriculture for crop monitoring, yield assessment and drought early warning. Soil-moisture deficits develop before vegetation stress appears in optical indices, so the SWI signals drought early. In general, the index can for instance be used to predict drought-related yield losses, as demonstrated for example for rainfed crops such as maize, sunflower and grapevine by Potopová et al. (2020). In another example, the World Bank used the SWI T=40 to quantify the welfare cost of drought across Sub-Saharan Africa, where soil moisture from SWI (along with NDVI) ranked among the best drought-hazard indicators (Gascoigne et al., 2024).

In hydrological application, the SWI can describe the wetness of the soil before a rainfall event, which controls how much rainfall becomes runoff. Assimilating it into a rainfall-runoff model improved runoff prediction, especially where the catchment's initial wetness was unknown (Brocca et al., 2010). It therefore can support flood forecasting and water-resource management.

Furthermore, in natural-hazard forecasting, soil wetness before rainfall affects slope stability. For example, adding the SWI to a rainfall-based predictor raised the correlation with landslide displacement from r = 0.40 to r = 0.85 in central Italy (Brocca et al., 2012).

The product is not suitable everywhere. Sections 5.1.2 and 5.1.3 describe the conditions where the SWI is less reliable.

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7 ANNEX: REQUIREMENTS

Following to the applicable document [AD1], the requirements relevant for SWI are described below.

7.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

SWI – Soil Water Index	
The surface soil water index provides daily information about moisture conditions in different soil depths. Soil moisture is a very heterogeneous variable and varies on small scales with soil properties and drainage patterns. Satellite measurements integrate over relative large-scale areas, with the presence of vegetation adding complexity to the interpretation. The SWI quantifies the amount of water (m^3/m^3) in soil layers at various depths and is derived from the SSM using a water balance model. The freeze and thaw status of soil is provided.	
Geometric Properties:	
spatial resolutions	3-10 km/1km
geolocation precision	Better than 0.5 pixels
coordinate position	Centre of the pixel
geodetical datum	WGS84
geographic projection	regular latitude/longitude grid
Geographic Coverage:	Global/EU
Time span of output product:	SWI: Daily SWI10: 10-day period SWI TS (timeseries): 6 months
Timeliness of output delivery:	SWI: 12hrs SWI-10: within 3 days after synthesis period
Uncertainty	Threshold: 0.1 m^3/m^3 Goal: 0.04 m^3/m^3
Stability	Threshold: 5% Goal: 3%

7.2 FURTHER REQUIREMENTS

7.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

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7.2.2 Data Structure

Data coding[8] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

7.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the CA during the FWC.

7.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS-245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

7.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

[8] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[9] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Land Surface Temperature (LST) and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

7.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.
The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[9] https://land.copernicus.eu/en/products/global-image-mosaic

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